

Context & Direction

This page will be discussing the direct research that lead to the design potentials I will be taking forward in this report

“A big part of the problem with components such as seats is that airlines don’t know what materials are in them,” says Tony Seville, Director of AIRA.

Seats in Storage

“Foams such as PU are banned from landfill due to Persistent Organic Pollutants (POPs), so instead they are combusted to take back some energy.” — Tony Seville

Further research shows that PU foam in airline seats is classified as a POP primarily due to the flame-retardant chemicals used in its manufacture. This means a whole seat cannot legally be sent to landfill unless it has first been disassembled so the foam can be removed. Because of this, large numbers of end-of-life seats are not disposed of, instead an estimated 1.5 million remain in storage worldwide.

From these stored seats, the aluminium content alone is approximately 14,400 tonnes, roughly the same mass as 10,000 cars. The seat frames are commonly manufactured from aluminium alloys such as 2024, 6082, and 7075, all of which are high-value alloys.

Disassembly Pains

Corrosion



Corrosion caused by mixing materials seized joints. This likely made safety checks redundant but also removes material value and extraction.

Tony Seville (AIRA) - Aircraft interior recycling specialist notes that AIRA only considers parts with up to 3% rust for recycling into the same material grade and will often send off aluminium parts with steel embedded into it to be recycled to a lower grade.

Standardisation

Having over 26 unique connectors required frequent tool changes. The connectors were in imperial units so metric tools did not fit in certain cases.



Joint Types



Some fasteners or joint connections were fidgety, things like grub screws mixed with adhesives almost broke tools and other connectors broke a hammer due to use of life influence preventing disassembly.

Adhesives



Usage of adhesives between materials made separation difficult and caused material contamination

It was often used to keep hook & loop fasteners attached to the structure which did help decrease time to remove covers and increased cover modularity

Time constraints

According to Tony Seville the whole seat needs to be dismantled in under 8 minutes for it to be economically viable

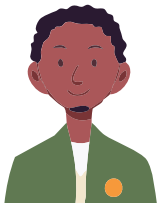
Industry Shortages

During Covid, to avoid bankruptcy, companies gave payouts to a significant number of workers before new workers were trained. This means currently in the industry there is a lack of skilled labour

Painted Fasteners

Painted fasteners are difficult to remove and the fastener material is completely unknown so it could contaminate sections.

Design Constraints



Name: Dave Donald
Occupation: Disassembler

- Pain Points:
- It is impossible to recycle components if we do not know what material is used.
 - Some parts of the seat cost more to disassemble than their worth, resulting in materials being scrapped.
 - We are expected to completely disassemble the seat in 8 minutes, but most of the time parts are not designed for quick disassembly.

- ☐ Disassembly must be quicker than 8 minutes
- ☐ The disassembly process must be low skill or easy to learn
- ☐ The process must be cheap so that it is economically beneficial for all parties
- ☐ A disassembler must be able to correctly identify materials

Design Directions

Directions gained from research

I need to design a seat that better suits the use phase environment, reducing corrosion to enable greater resource recovery and disassembly time. Potentially look into reducing moisture or looking further into mixing materials and the issues that causes.

My general structure should have standardised fasteners to decrease the number of tools needed intern reducing the time it takes to teardown. I should look at fasteners that still allow for structural integrity.

Joint types should be made less fidgety or awkward, the sequential order of the teardown should feel smooth and planned rather than messy

I could potentially find a replacement for adhesives, if not I can look at mitigating the effect they have in terms of contamination, at the same time the speed of the secondary structure deconstruction must be kept relatively constant.

I need to make it easier for companies to trace materials so that they can be easily sorted. This must be done with the lowest economic and IP risk so that companies are more likely to take this on

Problem Statement

The end of life disassembly of seats is often economically inefficient being that it is to expensive for the manual labour due to not being able to identify materials, seats having complex fastenings or due to use phase corrosion.

How might we design a seat for disassembly to reduce the skill and time required to disassemble creating a more economic and less wasteful system.

Traceability

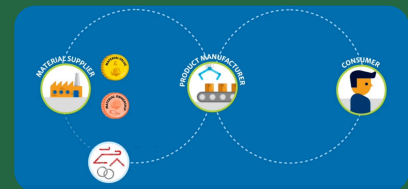
Material as a Service

This is the idea that a company loans its resources out, at its end of use the resources will return to the original company.

This is currently a theoretical concept that is attempting to gain traction



Imperial has had a major role in pushing materials as a service for years. They have worked with companies such as Ragn-Sells who are attempting to loan out chemicals used in water treatment.



This is a basic model of the concept created by imperial.

[1]

Positives

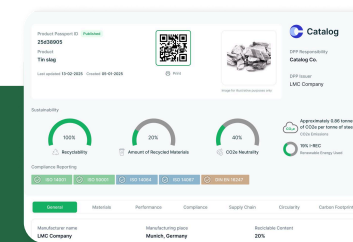
- This system reduces potential transport between stock holders
- Forces a closed loop circular system.
- IP doesn't have to be released at all since the supplier knows what materials are where already

Negatives

- The concept is experimental and therefore economically risky
- Due to the supply pyramid it seems unrealistic, and creates a large amount of resource debt that can last 5 to 10 years
- Logistics are difficult, specific components will have different suppliers and some parts can't be reused such as foams

Blockchains

A blockchain is a distributed database running on a peer-to-peer network, where each new block contains a cryptographic reference to the previous block. This creates a transparent and tamper-resistant chain of data.



MineSpider uses digital product passports to promote sustainable/responsible sourcing of materials and allowing companies to prove it.

They also track batteries from sourcing to use to end of life to help companies become compliant with EU regulations

[2]

Positives

- Allows for current power hierarchies in the industry to recycle without giving economic power
- Opens up an opportunity for smaller recycling companies creating a reverse pyramid at end of life
- Prevents green washing because everything has to be explicitly shown and cant be altered.
- Can scale globally across the supply chain, since it doesn't permanently trap materials in the system

Negatives

- Still requires big companies to take a certain amount of risk
- The cost of running the blockchain needs to be less than the potential profit it can bring which might take time to be realised
- Not necessarily a circular system or closed loop
- It only solves the issue if waste management companies exist
- Physical labels can be damaged

Chemical Tracing

Chemical tracing is when a material is marked with a unique chemical signature so it can be identified later.



Micro Trace is a company that allows people to track their products, they specialise in preventing counterfeit and false claims

[3]

Positives

- There is no physical label that can be damaged like the blockchain concept
- No digital infrastructure needed
- Can be tracked through multiple lifecycles through the system if digital infrastructure was added

Negatives

- Extremely expensive, needs equipment and experimental
- Detection and implementation machines must be purchased, its unlikely to happen and is a high upfront cost to joining the industry

Fasteners & Adhesives

Primary to primary connections

Painted Cross arm & Fastener



Aluminium 7075 does not share the same corrosive properties as lower grade aluminums. It is highly recommended that when in atmosphere it is painted. The issue with this is any fastener used is also painted and that material can not be identified.

[4]

Secondary to primary connections

Added Adhesives

Adhesives are often added during the use phase for things like repairs which can potentially ruin intended disassembly. Also many designs do not think much about the life jacket placement, and therefore adhesives are applied to the metal support shaft making it near impossible to pull of cross bars as intended.



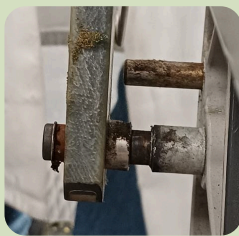
Rivets

The rivets in this image were used to attach one aluminium plate to another where the secondary plate had a adhesive with a hook & loop fastener connected. While it is a very small amount of material it would have to be cut off which when looking at the amount of seats adds up to a lot of material.



Poor Joint Choices

The joints & fasteners used to connect the primary to the secondary section were tedious & took way to long to be economically efficient.



Quick Fasteners

In reference to the table in the appendix Quarter turn and snap fastener seem promising to look at for secondary attachments. Also according to papers that discussed the fastest kind of fasteners, socket fasteners are the quickest to dismantle along with snap fasteners, taking less than 10 seconds. These fasteners are not strong enough for primary structure.

[5]



[a]

Detent socket Fastener

A quick-release joint where a pin with spring-loaded balls (or a spring ring) snaps into a matching socket. The detents expand into a groove inside the socket, creating a secure but tool-less connection that can be removed by pulling with moderate force.

Quarter Turn Fasteners

[6]



These fasteners latches inside a socket and remain locked until intentionally twisted. They are often used on high performing vehicles like planes, sports bikes etc to hold panels in place.

Snap on Fasteners

A joint that uses elastic features to deform and snap around another form. Easy to dismantle without tools, with good static strength and okay fatigue strength.

[15]

Primary Fasteners

Primary fasteners need to be stronger, this is due to the main stress load going through the primary frame.

The biggest issues while dismantling the primary structure are the bolts seizing and the secondary structure getting in the way.

Primary structure requires the use of bolts due to their superior load bearing, so this can not be changed.



[10]

What can be changed is the material. For example using the same material to the frame its connected to e.g. 7075 bolts, this creates issues as to whether the material can bear the load needed. Some materials such as SMAs could be used to quickly release the bolts.

Another response might be creating a system to pull out the bolts quicker e.g. one tool for all the bolts.

Secondary Requirements

- ☐ This section of the seat is worth less economically so has to be quick to disassemble
- ☐ If it is easy to disassemble it needs to be hidden from a general seat user
- ☐ If obstructing primary structure it is more economic to destructively disassemble and it should be in this case as it means the rest of the chair is more likely to be recycled due to increased value.

Primary Requirements

- ☐ It needs to be strong, the primary structure holds most of the stress
- ☐ If painted it either needs a method to disassemble or to be the same material as the surroundings

Overview

Business Model

Product Passport

Scan the QR code on your airline seat to view detailed product information

Scan QR Code

Point your device at the QR code or tap the button above

Digital Product Passport

Material Composition

Aluminum components	Grade 6061-T6	35%
Plastic components	Grade 1 (PE-400-40)	15%
Carbon Fiber TUBES	Aluminum Grade	25%
Plastic SHEET	PLAS 105	15%
Steel Reinforcement	Steel Reinforcement Grade	10%

Product Passport application

A disassembler would be able to scan a qr code. If they have access to the blockchain they will be given access to the details of the seat. They will see what materials are used, where they are used and if parts disassembly is less obvious, for example the SMA Rivets idea, how they might disassemble it.

Business application

Companies with more power in the industry would give access to the blockchain to recycling companies such as AIRA, allowing for a privatised circular economy because resources are easier to identify and less seats would go to warehouses since the value of the product increases.

Fasteners, Joints & Adhesives

SMA Rivets	SMA's are materials that change shape when heated above a set activation temperature. When used as rivets, they behave like normal fasteners during service but contract or release when heated with an electric clamp.	SMA would allow rivets to be removed quickly and cleanly without damaging the surrounding structure. Improving overall disassembly speed and reducing material contamination.
Snap on adhesive sacrifices	Snap-on adhesive sacrifices are flexible polymer clips that attach to the seat frame and provide a surface for adhesives or Velcro. They prevent glue from touching the aluminium and can be pulled off by hand during disassembly, leaving the primary structure clean and easy to recycle	This would remove the need for saws and other tools to take the main frame apart due to glue making disassembly difficult. It would also reduce contamination in general due to adhesives increasing the value of the chair as more material can be gathered and sold.
Arm rest design	The idea of this design is to use a detent rail with quarter turn fasteners. The quarter turn fasteners would release the rail when the arm rest is at its max angle. This would allow it to be detached	This improvement would allow disassemblers to quickly remove the secondary frame from the primary frame, with the least amount of tool use possible
Overhaul	This would include basic bolt material changes, moving away from possibly corrosive materials that might cause seizing to materials that hold the same structural integrity, are similarly dense and corrosion resistant.	Reduces the chance of running into seized joints. This drastically improves disassembly time as more advanced tools like angle grinders/saws are not needed.

Design Basics

This is a discussion of the parts of the seat that are not the direct focus of report but still important in terms of the actual seat, mostly includes secondary structure

Dress Coverings

The seats dress coverings will be made from Gen Phoenix's ELeather.

[8]

They have a take back system like a MaaS or PaaS at the end of life adding the circularity of the design.

Despite spacer fabrics having better thermal comfort than ELeather, it is less desirable in terms of recyclability since it has a mix of materials and this project is focusing on end of life therefore ELeather is the superior choice.

[7]

Seat Foams

Future foams produces PU:

- ZHR24270
- ZHR70280

These are both **different materials** which should be against the disassembly goals but since **its PU with POP fire retardants** they **will both be incinerated** at end of life and therefore it doesn't matter.

ZHR70280: This material has a higher Indent Force Deflection meaning its **significantly firmer**, it will be used for the **main seat cushion** due to its **greater load bearing** capabilities.

ZHR24270: This material will be used for **seat backs** since it is **softer** and more comfortable.

[Read Appendix]

LOPA

Wide Body

Narrow Body

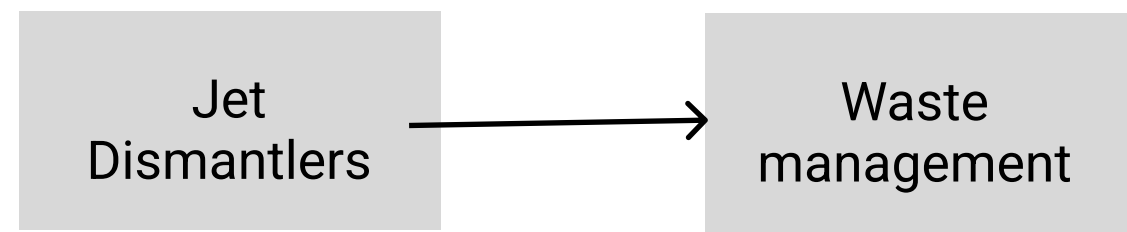
I will not be changing the LOPA, there might potentially be benefits in doing so in terms of removing seats and the general interior when the seats reach their warranty end but this stage of the process does not have much public information.

[7]

Business Model

Current System

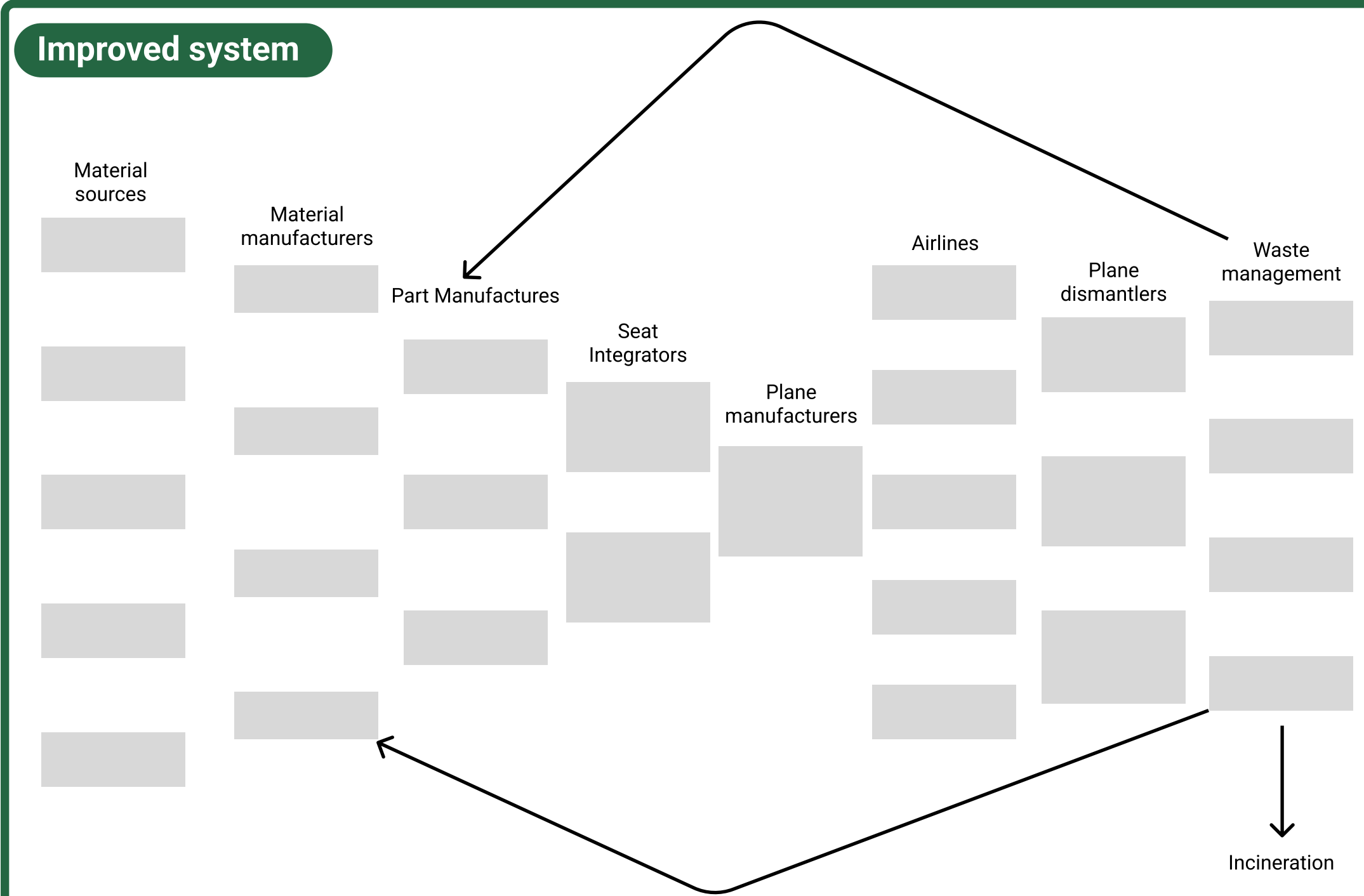
In the current system there is an oligopoly:
5 seat integrator companies own 55% of the market and 2
jet manufacturing companies own 80% of their market.



Due to their general need to keep IPs secret there is almost no waste management companies other than AIRA meaning there is no competition in the waste management side of the industry

The lack of competition reduces waste management innovation because stable companies do not want to take risk.

Improved system



Box size = system influence, Box count = how centralised that part of the industry is

System Benefits

The Barcode system would allow larger companies to send seats off without giving IP. The Seat integrators would be able give waste management companies access to the blockchain. This would allow waste management companies to differentiate between grades of materials.

One major improvement would be the separation of 7075 from the other alloys.

According to Tony Seville 7075 can be sold at £150 per kg while mixed aluminium is usually around £1000 per ton.

Due to the improvement of relationships (Waste management are making higher profits) they are likely to sell back into the industry. Potentially at lower prices compared to virgin material. This would mean both waste management companies and the big industry players would both benefit.

Privacy would be kept while allowing access to more competition within waste management industry

Seat End of Life

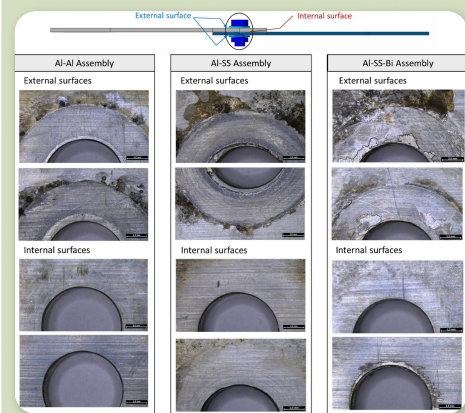
The use of this model, with my seat design or not drastically improves the state of the end of life. Since companies can identify materials, it becomes more profitable to disassemble them than bare the responsibility of storing them. Less material will be wasted due to the increased economic incentive to sell materials.

Since more money can be made from the alloys and therefore individual chairs hold more economic value. The cost of disposing of the air filled PU is reduced, and there is the potential of bulk PU being profitable if sent to incineration or shredded and reused for other means.

Material Analysis

Al--Steel usage

Currently there is a high usage of mixing Al alloys with Stainless steel fasteners.

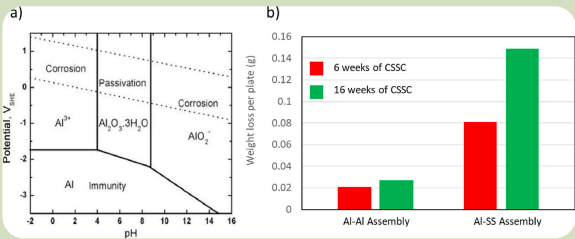


Al is often known for its self protecting layer against corrosion. This mostly applies to lower grade aluminium. Higher grade aluminium is known to be self destructive with internal galvanic corrosion between phases. They trade corrosive resistance for strength. Aluminium is also quite low on the galvanic chart meaning it will mostly act as a sacrificial anode.

[11]

These tables clearly show the corrosive affects of Mixing Al and SS and its affects on weight loss

[12]



Steel Properties

Steel is used for one obvious reason, its cheap while still holding a high yield strength.

For comparison if we compare 7075 AL to steel 4140 yield strengths, the aluminium sits at around 502Mpa while 4140 sits between 600-900Mpa

Steel is also significantly more ductile with an elastic modulus of around 200GPa while 7075 sits at 71-72Gpa.

One downside of steel is its density sitting at around 7850kg/m^3 while 7075 sits at 2810kg/m^3.

[13] [14]

Material Requirements

These requirements for my material choice are based on the issues and benefits of steel.

- ☐ Be more galvanically compatible
- ☐ High strength
- ☐ High ductility
- ☐ High environmental corrosion resistance.
- ☐ The bolts should sit at a density of 7850 or less.

Potential SMA materials

CuZnAl [19] [20]
Density: 7600 Kg/m^3
Elastic modulus: 92 Gpa
Yield strength: 590 Mpa
Corrosive Resistance: High

SMA transformation
35 - 70 degrees celcius

Sample	Mn (K)	Mo (K)	Al (K)	Al (K)
Cu-Zn-Al alloy	320	307	328	343

Comparison to Steel:
It is close to lower grades of 4140 but does not live up to higher in terms of strength and ductility.

Replacement Rating: ★★ ★

NiTi [16] [17][18]
Density: 6500 Kg/m^3
Elastic modulus: 41-75 Gpa
Superelastic Plateau stress : 450 Mpa
Corrosive Resistance: High nobility but when it does corrode its toxic, and it corrodes to bodily fluids.

SMA transformation
are negative to room temp

Comparison to Steel:
Not a good replacement, significantly weaker and potentially toxic.

Replacement Rating: ★

FeMnSi [21] [22]
Density: 7200 Kg/m^3
Elastic modulus: 170 Gpa
Yield strength: 200 - 300Mpa, there are recent papers that suggest this can be increased through crystal orientation.
Corrosive Resistance: Medium

Comparison to Steel:
Other than yield strength this material is perfect unfortunately the yield strength is important, this material is promising for the future.

Replacement Rating: ★★

SMA Summary

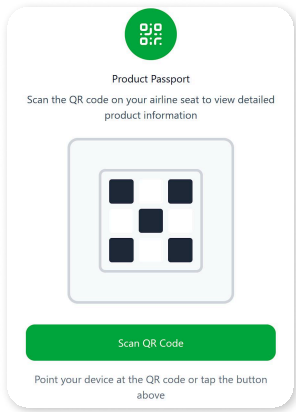
Summary
Unfortunately none of the materials are a 100% great replacement for steel. They are all noticeably more noble than aluminium, and as discussed previously higher grade aluminiums suffer more from corrosion. This issue still exists for steel so its not that big of an issue.

CuZnAl
This alloy has significantly less ductility but close performance in terms of strength, this could be used for higher stressed joints but would need to be tested to see if it meets policy standard.

FeMnSi
This is a great replacement for fasteners that suffer less stress, for example the rivets that attach secondary components, this might be useful for attachment of my snap on sacrifices.

Final Design

Material Identification



Dave is able to scan the qr code that has been burnt into the underside frame via a laser annealing. If he has access to the blockchain this tells him what material each part of the seat is made of and instructions on how to dismantle.

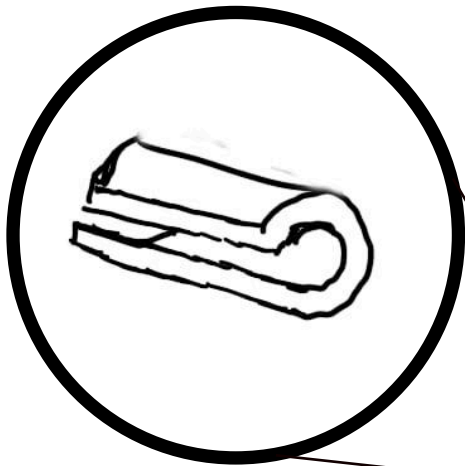
Secondary components



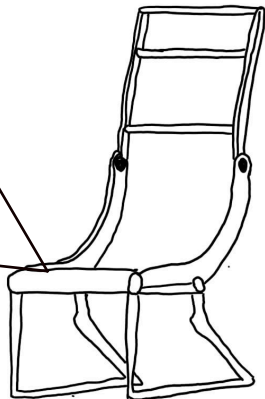
The dress cover is designed to be removed via Velcro and the foam will snap on to the primary frame.

The Velcro will be attached to sacrificial plastic that is riveted into the aluminium seat back frame with FeMnSi SMA, this is so it can be quickly removed for quick access to the aluminium back frame.

Sacrificial snap on



Sacrificial snap on that the adhesive for the life jacket will go on. Dave can pull this off with his hands.



Stronger Bolts

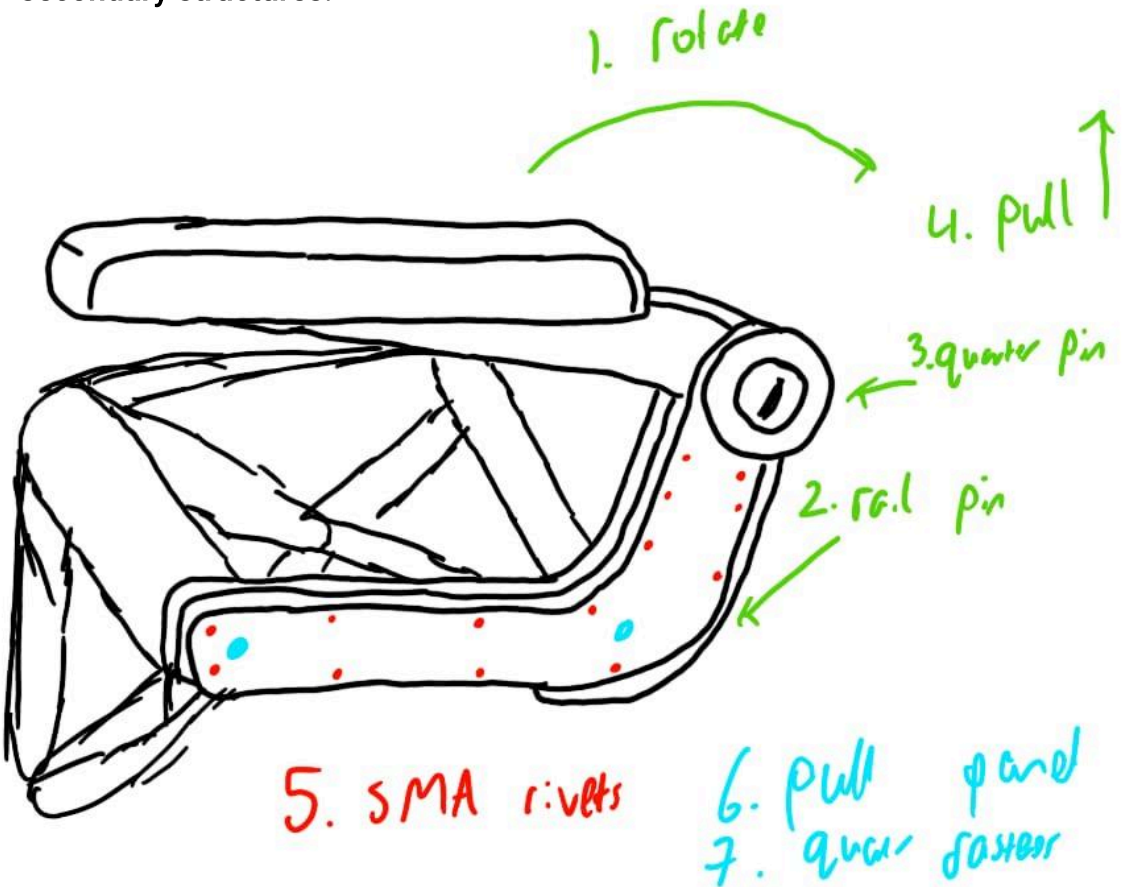
The bolts that need more strength will be made from CuZnAl, this will need to be tested to see if the bolts reach policy regulations and passes the 16g tests. If they pass the tests these will be used to quickly remove bolts from high grade aluminium using an electric clamp

Arm rest Design

The snap on PU foam would be removed as well as the dress covering.

1. The arm rest is rotated up
2. There is a rail pin behind the arm rest previously hidden by the dress cover and foam.
3. The quarter pin on the joint is removed
 - a. the centre seat will have a quarter pin as well, this will sit on the back of the rest when removed it will allow the back seat to be removed.
4. Pull the detent rail out.
5. Use electric clamp to remove panel rivets and then pull panel off
6. quarter fasteners turned
7. pull arm rest off.

Now disassembler Dave has full access to the primary structure and it is separate from the secondary structures.



Disassembly Tree Comparison

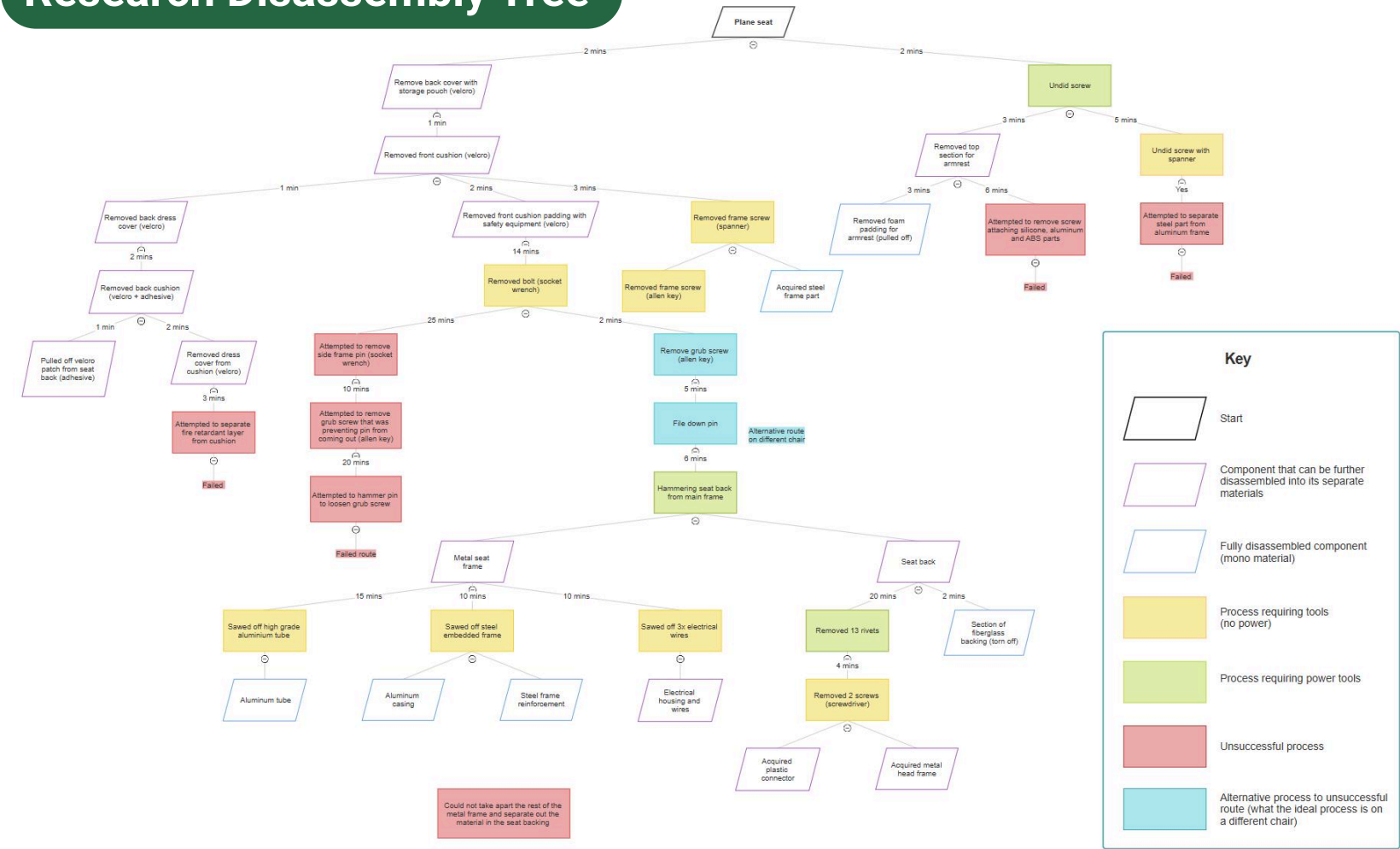
Designed Teardown

The actual time taken between each step would need to be measured before it can be compared fully



Comparison

Research Disassembly Tree



Total Time = 3hrs with 3 people

The obvious difference between my disassembly design and the researched tree is the linearity. With the blockchain qr code there will be instructions on how to disassemble rather than just figuring it out.

The next major difference is the tool usage, one of the requirements was to reduce the tool usage. The only part of the disassembly that should require a tool is the SMA bolts and rivets, this would use an electric clamp to trigger the transformation phase, the rest of the teardown is built on a sequence of quarter turns and snap ons.



Due to the lack of material identification in the research dismantling we ended up with a large amount of mixed alloys. With the material identification system this would instead be sorted into specific materials.



Name: Dave Donald
Occupation: Disassembler

Pain Points:

- It is impossible to recycle components if we do not know what material is used.
- Some parts of the seat cost more to disassemble than their worth, resulting in materials being scrapped.
- We are expected to completely disassemble the seat in 8 minutes, but most of the time parts are not designed for quick disassembly.

Design Constraints

- ☐ Disassembly must be quicker than 8 minutes
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Appendix

PU foams data sheet

[24]

FUTURE FOAM, INC
PRODUCT DATA

PRODUCT: ZHR70280

PRODUCTION FACILITY: Co. Bluffs, IA

DATE: 2021

<u>PHYSICAL PROPERTY</u>	<u>VALUE</u>	<u>UNITS</u>	<u>TEST METHOD</u>
Color:	Natural (white)		visual
Density:	2.70 - 2.90 lbs. / cu.ft.		ASTM D-3574-17
Indent. Force Deflection @ 25% deflection:	63.0 - 77.0 lbs. / 50 sq. in.		ASTM D-3574-17 (4" Specimen)
Tensile Strength:	>10.0 lbs./sq.in.		ASTM D-3574-17
Elongation:	>100 %		ASTM D-3574-17
Tear Resistance:	>1.0 lbs. / lin. inch		ASTM D-3574-17
Compression Set @ 50% comp.:	<10.0 %		ASTM D-3574-17
Flammability:	Pass		CA TB 117-2000

This open celled flexible polyurethane foam has an HR prefix which designates this as a High Resilience product. The cell structure and specialty raw materials used in this product provide higher resilience than conventional foam. The Z prefix indicates this product contains flame retardant as defined by SB1019

This product has been found to meet the test requirements of Section 3 of the California Technical Bulletin #117-2013.

FUTURE FOAM, INC.
PRODUCT DATA

PRODUCT: ZHR24270

PRODUCTION FACILITY: Co. Bluffs, LA

DATE: 2021

<u>PHYSICAL PROPERTY</u>	<u>VALUE</u>	<u>UNITS</u>	<u>TEST METHOD</u>
Color:	Natural (white)		visual
Density:	2.60-2.80 lbs. / cu.ft.		ASTM D-3574-1
Indent. Force Deflection @ 25% deflection:	21.0 - 30.0 lbs. / 50 sq. in.		ASTM D-3574-1 (4" Specimen)
Tensile Strength:	>10.0 lbs./sq.in.		ASTM D-3574-1
Elongation:	>100 %		ASTM D-3574-1
Tear Resistance:	>1.0 lbs. / lin. inch		ASTM D-3574-1
Compression Set @ 50% comp.:	<10.0 %		ASTM D-3574-1
Flammability:	Pass		CA TB117-2000

Fastener rating chart

[23]

		Material Connection		Fictional Connection				Positive Connection				
		Adhesive Bonding	Welding	Magnetic	Velcro	Nut & Bolt (metal)	Spring Connection	Snap	Bent Lever	¼ turn	Press-turn	Press-press
Carrying Capacity	Static Strength	2	1	2	3	1	2	1	1	1	2	2
	Fatigue Strength	2	1	2	3	1	3	2	2	2	2	3
Joining Behaviour	Joining Expenditure	2	2	1	1	2	1	1	1	1	1	1
	Guidance Expenditure	3	3	2	1	2	1	1	2	1	1	1
Detaching Behaviour	Detaching Expenditure	3	3	1	1	2	1	3	1	1	1	1
	Destructive Detaching Expenditure	2	2	N/A	N/A	2	N/A	1	2	2	2	2
Recyclability	Product Recycling	3	3	2	2	2	1	3	1	1	1	1
	Material Recycling	2	1	2	2	2	1	1	1	1	2	2

1 = Good 2 = Average 3 = Poor

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